

Measuring a Determinability Boundary in Bits

A Certified Simulation of the Event Density Substrate

Allen Proxmire

June 2026

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Architectural Distillation corpus · evaluations/ED_Substrate/Bits/

Abstract

The Event Density (ED) framework models physical reality as a discrete substrate of micro-events that influence one another only across a finite reach, bounded by *decoupling surfaces*. A prior structural evaluation established that these surfaces constitute a **determinability boundary**: a horizon past which one region of the substrate cannot, even in principle, determine the state of another. That result was qualitative. This paper makes it quantitative. We build a faithful executable model of the

ED substrate — discrete stability-maximizing selection under irreversible commitment, *not* a coarse-grained approximation — and subject it to four pre-registered correctness gates (monotonicity, acyclicity, factorization, unique selection), all of which it passes (20/20 automated tests). On a certified two-region test substrate separated by a single decoupling surface, we then measure the **determinability-boundary contrast** Δ in bits: the mutual information that propagates *within* a reach-connected region minus the mutual information that crosses the boundary. We find a positive within-region contrast with the across-boundary term indistinguishable from a shuffle null. Subsequent robustness sweeps (§11) sharpen the result decisively: the contrast’s *magnitude* is observable-dependent (ranging 0.17–0.98 bits across readouts of the same states), so there is **no single intrinsic scalar** for ED — but across-boundary mutual information $M_2 \approx 0$ holds invariantly across substrate size, estimator, and observable. The robust finding is therefore structural, not numerical: the boundary’s **perfect, observable-independent severance** is what characterizes it. We argue this demonstrates a general method (a finite-reach determinability boundary is measurable) while emphasizing strict limits: the work is structural, not physical; and a parallel to the parity barrier of multiplicative number theory is asserted at the level of *form only* — a shared boundary-that-destroys-determination, **not** a shared scalar — with no cross-domain content claim.

Keywords: Event Density, determinability boundary, finite reach, mutual information, factorization, discrete substrate, Architectural Distillation.

1. Introduction

A recurring idea across several formal systems is that of an architecture with **finite reach**: a generative structure whose elements influence one another only out to a bounded horizon. Where reach is finite, a natural consequence follows — there exists a boundary past which the architecture cannot determine what lies beyond. We call this a **determinability boundary**.

In the Event Density (ED) framework, finite reach is built in at the axiom level through *decoupling surfaces*, beyond which reciprocal participation between regions ceases. A structural evaluation of the ED substrate (the twelve-document analysis preceding this work) concluded that **the determinability boundary of ED is exactly the decoupling surface**: inside a reach horizon the substrate is determinable from local information; across it, no channel carries the determining information. The reach horizon and the determinability horizon coincide.

That conclusion was reached structurally — by analysis of the axioms, the update rule, and the constraint geometry. It is the kind of claim that invites a measurement. If determination genuinely fails across a boundary, then information about one side, given the other, should be *quantifiably* reduced — and information theory provides the natural currency: **bits**.

This paper reports that measurement. We construct a certified simulation of the ED substrate and measure the bits of determination that the decoupling surface blocks. The contribution is twofold: (i) a *method* — a reproducible, gated procedure for measuring a determinability boundary in bits; and (ii) a *result* — a positive, stable measurement on a controlled test substrate, with the information-theoretic controls behaving exactly as the structural theory predicts.

We are deliberate about scope. The number we report is a property of our test configuration, not of ED as a whole; the work concerns *structural* well-builtness, not physical truth; and a parallel we draw to a boundary in number theory is one of architectural form, never of content. These limits are stated where they arise and consolidated in §8.

2. Background

2.1 The Event Density substrate

ED posits a discrete relational substrate with the following primitive commitments (the substrate's axioms, labelled B1–B7 in the source evaluation):

- **Micro-events (B1)** — atomic, indivisible acts of becoming; their local rate is the *event density*.
- **Channels (B2)** — participation relations on stable subgraphs, each weighted by a bandwidth.
- **Adjacency and chains (B3)** — bandwidth-gated adjacency; sequences of adjacent events form *chains* (which coarse-grain to particles).
- **Commitment density ρ (B4)** — the local accumulation of committed events; the substrate's state variable.
- **Orientation (B5)** — a relational directionality carried by the channel structure.
- **Decoupling surfaces (B6)** — horizons of finite reach, where reciprocal participation between regions becomes one-sided.
- **Commitment irreversibility, “P11” (B7)** — every commitment, once made, is never unmade; evolution is forward-only.

The substrate evolves by an explicit, local **selection rule**. A chain at a position evaluates each admissible next event e' by a stability score

$$\Sigma(e') = \text{Coh}(e', s) - \text{Str}(e', \rho_{\text{local}}) - \text{Grad}(e', \nabla \rho),$$

where Coh rewards coherence with the current chain state, Str penalizes participation strain against the local commitment density, and Grad penalizes fighting the local density gradient. The chain commits the **maximum- Σ** candidate. This is a discrete, local, *selection* law — it picks an extreme, it does not average.

A point of emphasis that proves important below: ED's familiar continuum description is a degenerate-parabolic (diffusive) partial differential equation. That diffusive law is the **coarse-grained shadow** of the substrate — what one obtains by averaging the selection rule over many chains and many steps. The substrate itself, being maximization-driven, is *sharpening*, not diffusing. The diffusion is an artifact of zooming out, not a property of the substrate (see §7).

2.2 The determinability boundary

A **reach stratum** is a maximal set of micro-events mutually reachable without crossing a decoupling surface — formally, a connected component of the *reciprocal* (non-decoupled) participation subgraph. Because reach is finite (B6), the substrate partitions into such strata, and the reachable configurations *factorize* across them: regions separated by a decoupling surface are causally independent. One-sided influence may cross a surface, but reciprocal participation — the kind that would let one region determine another — does not.

The structural evaluation's central finding is that this factorization *is* the determinability boundary. Within a stratum the substrate is determinable from local channels; across a decoupling surface, no channel carries the determining information. The boundary is therefore a positive structural feature (an axiom, B6), not an absence — and it is the natural object to quantify.

2.3 A structural parallel — stated as form only

The motivation to measure ED’s boundary in bits comes partly from a parallel with an unrelated system. Separate work on the multiplicative structure of the integers (the “Factor Skyline” programme) identifies its own determinability limit — the **parity barrier** (a Sarnak-disjointness phenomenon) — and has *quantified* it (an “escape density” of roughly 0.265 in its own decomposition). The two systems share an architectural *shape*: a finite-reach generative substrate, a determinability boundary at the reach horizon, and a smoother continuous projection.

We use this parallel only to motivate the *kind* of measurement, and we state the discipline plainly and keep it throughout: **the analogy is structural, detected by a shared evaluation methodology; it is not a claim that ED explains primes, that the prime work explains physics, or that the two boundaries are the same object.** Placing ED’s measured number beside the parity barrier’s is a comparison of *roles*, never of *values*.

2.4 Why a custom simulator

Two existing ED codebases were surveyed. Both are valuable but neither suffices: they evolve by diffusion / gradient-flow dynamics, which is precisely the *shadow* behaviour (averaging), not the *substrate* behaviour (selection). Measuring the substrate’s determinability boundary requires the substrate’s own sharpening, selection-driven rule. We therefore built a minimal, faithful simulator implementing Σ -maximization directly, reusing only general infrastructure (information-theoretic estimators) from the surveyed code.

3. The measurement question

We define the **determinability-boundary contrast** as the difference between predictability within a stratum and predictability across the boundary, in bits:

$$\Delta = M_1 - M_2$$

where, with all terms computed as mutual information (MI) over an ensemble of independent runs:

- M_1 = **within-stratum** predictability — MI between two reach-connected halves of a single stratum;
- M_2 = **across-boundary** predictability — MI between two reach-decoupled strata;
- M_3 = **shuffle/null control** — MI between one stratum and a randomly permuted copy of the other, giving the estimator’s finite-sample floor.

The theoretical prediction is sharp: reach-connected regions share information ($M_1 > 0$); reach-decoupled regions do not ($M_2 \approx 0$); and M_2 should be indistinguishable from the null floor M_3 . The contrast Δ is then the positive quantity of determination that lives within a stratum but does not cross the boundary — *the bits the boundary blocks*.

Crucially, M_1 , M_2 , and M_3 are computed with the **same estimator and the same binning**, so $\Delta = M_1 - M_2$ differences out the estimator’s shared finite-sample bias. This matched-difference design is what makes a small absolute number trustworthy.

4. Methods

4.1 Simulator

The simulator is a seven-module Python package implementing the substrate faithfully:

- a **participation graph** with bandwidth-weighted channels and a per-edge decoupling flag;
- **node state** carrying commitment density ρ and orientation, with a single write path — `commit()` — that enforces irreversibility (it rejects any decrement of ρ);
- the Σ **functional**, computed from ρ and graph-local structure only (it is, by construction, blind to orientation — orientation contributes no stability score, consistent with the substrate’s specification);
- a **deterministic tie-break**: among Σ -maximal candidates, the winner is the one with lexicographically maximal key (bandwidth, node id) — a total order, so the next step is always unique;
- the **update loop**, processing active chains in a canonical (stratum, node) order with commitments immediately visible, making the whole evolution a deterministic function of (graph, initial state, parameters);
- **reach-stratum detection** (connected components of the reciprocal subgraph) and **decoupling-surface detection** (the boundary edges and the across-boundary node pairings);
- an append-only **recorder** that captures per-step ρ /orientation and the commitment log, and exports them.

Randomness enters *only* through the seeded initial conditions; trajectories themselves are deterministic — the ensemble model the measurement requires.

4.2 Test substrate (the controlled boundary)

The measurement substrate is two isomorphic eight-node chains, $A = \{0, \dots, 7\}$ and $B = \{8, \dots, 15\}$, joined by a single **decoupled bridge** (7, 8). The reciprocal subgraph thus has exactly two components — two reach strata — separated by one determinability boundary. Each chain is seeded from its **own independent** random initial conditions (distinct seeds), so B is a genuinely independent draw from A . Evolution runs to a natural fixed point (no admissible positive-score move remains) or a step cap; in practice it reaches the fixed point quickly.

4.3 Ensemble and estimator

An ensemble of N independent runs is generated with independent per-stratum seeds. For each run we extract summary statistics (summed final ρ) for: stratum A , stratum B , and the two reach-connected halves of A . Mutual information is estimated with a **Miller-Madow-corrected** histogram estimator (quantile-edged bins), and converted from nats to bits ($1 \text{ nat} = 1/\ln 2 \text{ bits}$). The Miller-Madow correction is necessary because the naive plug-in MI is *positively* biased at finite sample size: truly independent variables yield a small spurious positive MI without it. We report $N = 200$ as the primary ensemble, with $N \in \{100, 200, 400\}$ for stability.

4.4 Reproducibility by construction

Every ensemble uses fixed seed bases, so the entire measurement reproduces exactly. The simulator’s determinism is itself tested (identical inputs yield bit-identical histories).

5. Verification: the four correctness gates

No determinability number is measured until the simulator is **certified** against four pre-registered gates. The gates were fixed in advance, as general criteria, before the measurement substrate was scored against them — the discipline that prevents tuning the instrument to a flattering result.

Gate	Requirement	How verified
1 — Monotonicity	ρ never decreases	Recorded history is elementwise non-decreasing; <code>commit()</code> rejects negative increments.
2 — Acyclicity	No global state recurs	Every per-step state signature is unique; total ρ strictly increases.
3 — Factorization	Decoupling severs reciprocal participation	<i>Structural</i> : perturbing one stratum's seed leaves the other byte-identical, and no commitment crosses the boundary. <i>Informational</i> : across-boundary MI ≈ 0 , matching its shuffle control.
4 — Tie-break	Unique, deterministic selection	Forced ties resolve to exactly one candidate by the lexicographic key; orientation flips leave Σ unchanged.

Gate 3 is the load-bearing one and is tested two independent ways. The structural test is *exact*: because the strata are causally independent, changing B 's random seed must leave A 's entire commitment history and final state **bit-for-bit identical** — and it does, with the symmetric statement for B under an A -perturbation. If a single channel leaked across the boundary, this exact equality would fail. The informational test corroborates: the mutual information between the two strata is statistically indistinguishable from zero and from a shuffle null.

A consolidated certification harness runs all gates together. The result:

```

pytest          : 20 passed, 0 failed
Gate 1 mono     : PASS
Gate 2 acyclic  : PASS
Gate 3 factor   : PASS   (MI = -0.0017 nats; exact A<->B independence)
Gate 4 tiebreak : PASS
SIMULATOR: CERTIFIED - Delta measurement permitted.
```

One verification subtlety, recorded for transparency: an early acyclicity check flagged a recurrence that was, on inspection, the terminal fixed point recorded twice (a stationary endpoint, not a cycle). The recorder was corrected to log a snapshot only when the step changed the state; the genuine invariant — total ρ strictly increasing on every recorded step — holds exactly. This is stationarity, not a violation of acyclicity.

6. Results

With the simulator certified, the determinability-boundary contrast was measured on the test substrate. The primary ensemble ($N = 200$, 2-bin Miller-Madow estimator, bits):

Quantity	Definition	Value (bits)
M_1 — within-stratum	$\text{MI}(A_{\text{left}}; A_{\text{right}})$	+0.092
M_2 — across-boundary	$\text{MI}(A; B)$	−0.002
M_3 — shuffle null	$\text{MI}(A; \text{shuffled } B)$	−0.001
$\Delta = M_1 - M_2$	determinability-boundary contrast	+0.094

The residual $M_2 - M_3 = -0.0014$ bits confirms the across-boundary term sits at the estimator’s null floor. Stability across ensemble sizes:

N	M_1	M_2	Δ
100	+0.0884	+0.0032	+0.0852
200	+0.0920	−0.0025	+0.0945
400	+0.0833	+0.0000	+0.0833

The spread in Δ over $N \in \{100, 200, 400\}$ is 0.011 bits — an order of magnitude below the signal.

Reading. The three measures behave exactly as the structural theory requires. Within a reach stratum, the two halves share real information ($M_1 \approx 0.09$ bits). Across the determinability boundary, shared information collapses to the noise floor ($M_2 \approx M_3 \approx 0$). The contrast

$$\Delta \approx 0.09 \text{ bits}$$

is positive, well above the ~ 0.001 -bit floor, and stable. **The determinability boundary blocks a real, repeatable, measurable quantity of information.**

7. Discussion

7.1 What the number means: a picture

It is easy to misread $\Delta \approx 0.09$ bits as “the most a single event can hand off.” It is not that. Δ is a **contrast between two correlations**, measured across the ensemble — and the intuition is best built through a concrete frame.

The betting frame. Run the substrate many times. Each run, record how “busy” (high- ρ) the **left half** of region A ended up, and how busy its **right half** ended up. Now ask: *told the left half was busy, how well can you bet on the right half?*

- *Within a region (the two halves of A).* A little better than chance. The halves belong to one reach-connected region, so a faint shared signal links them — told “left busy,” you would bet “right busy” and win modestly more than half the time. That faint edge is $M_1 \approx 0.092$ bits.

- *Across the boundary (region A vs. region B).* Not at all. Told everything about A, your best guess about B is an even coin flip. The two strata are statistical strangers. That is $M_2 \approx 0$ bits.

Δ is the *betting edge you have inside a region minus the edge you have across the boundary*: $0.092 - 0 \approx 0.09$ bits.

The soundproof-wall frame. Inside region A, two halves can faintly hear each other through a thin wall. The decoupling surface is a **perfectly soundproof wall**: whatever happens in A, region B hears nothing. Δ measures *how much hearing the wall removes* — the faint within-region signal (≈ 0.09 bits) that the boundary erases to silence (0 bits).

The scale of a bit. One full bit of mutual information is a perfect yes/no link (tell me X, I know Y for certain); zero bits is total independence. 0.09 bits is therefore *small* — a faint lean, not a lock. Inside a region: a weak hint. Across the boundary: no hint at all, indistinguishable from randomly shuffled data.

The contrast is the result, not the value. The scientifically meaningful content is not the magnitude 0.09 but the *pattern*: **nonzero within a region, exactly zero across the boundary**. The zero is not trivial — there genuinely is a signal to block (the nonzero M_1), and the boundary blocks all of it, down to the noise floor. That is the determinability boundary rendered numerically: knowability lives inside reach and dies at the boundary. The value 0.09 is merely *how large the erased signal happened to be in this test configuration* (see §8.1); the erasure itself — the collapse from a real signal to zero — is the finding.

This also clarifies the comparison to the parity barrier (§2.3): that programme's ≈ 0.265 is *domain-intrinsic* — a fraction of multiplicative structure that genuinely escapes determinability, a property of the integers themselves. The present 0.09 is a *configuration baseline*, not yet an intrinsic ED quantity. The two are alike in *kind* (a determinability boundary measured in bits), not in being comparable values — exactly the role-only parallel maintained throughout.

7.2 The structural claim, now quantified

The structural evaluation argued that ED's decoupling surface is a determinability boundary. This measurement gives that claim a number. The qualitative prediction — information propagates within a reach horizon and stops at it — is borne out quantitatively: $M_1 > 0$, $M_2 \approx 0$, and the gap Δ is positive and stable. The boundary is not a turn of phrase; it is a measurable information barrier.

7.3 Substrate sharpens; shadow diffuses

A result that emerged from building the faithful simulator deserves emphasis. ED's continuum shadow is diffusive (a degenerate-parabolic PDE), but the substrate, driven by maximization, *sharpens*. The diffusion is **manufactured by coarse-graining** — averaging a sharpening selection rule over many events — and is absent from the substrate itself. This is why a faithful measurement could not have used the diffusive shadow: it would have measured the wrong object. The simulator implements the selection rule directly, and the certified factorization is a property of that rule, not of any averaged approximation. More broadly, this shows that a system's behaviour at the substrate level need not match its behaviour at the coarse-grained level — a methodological caution of independent interest.

7.4 The method generalizes

Nothing in the measurement procedure is specific to ED's particular parameters. The construction — partition a finite-reach substrate into strata, run an ensemble, contrast within-stratum against across-boundary mutual information with a matched estimator — applies to *any* discrete relational substrate with a determinability boundary. ED is the first worked example; the procedure is the transferable contribution.

8. Scope and limitations

We state the bounds of the result explicitly.

1. **The magnitude is configuration-specific.** $\Delta \approx 0.09$ bits is a property of this test substrate, not a constant of ED. It depends on the summary statistic, the binning, the Σ coefficients, the extinction threshold, the topology, and the seeding. What is robust is the *pattern* ($\Delta > 0$, $M_2 \approx M_3$, $M_1 > M_2$); the *value* is not yet a property of the architecture. Establishing a configuration-independent ED value would require a systematic robustness program (varying topology, reach profile, coefficients, and summary), which is research-scale and separate from this proof-of-method.
 2. **This is structural, not physical.** The entire programme — both the prior evaluation and this measurement — concerns *architectural well-builtness*. It does not test, support, or undermine ED as a description of the physical world. A boundary being measurable in bits is a fact about the architecture, not about nature.
 3. **The number-theory parallel is form, not content.** The comparison to the parity barrier is role-to-role: both are finite-reach determinability boundaries admitting a bits-denominated measure. We make **no** claim that 0.09 and the parity barrier's ~ 0.265 are comparable in value, that ED's boundary *is* a parity barrier, or that the domains share subject matter.
 4. **Small substrate, simple estimator.** The test case is sixteen nodes with a single constructed boundary, and the MI estimator is a 2-bin histogram with bias correction. These suffice for a matched-difference proof-of-method but are not a high-resolution measurement; larger substrates and richer estimators are part of the robustness program.
-

9. Reproducibility

The complete simulator, analysis, tests, and step-by-step record are in `evaluations/ED_Substrate/Bits/`. To reproduce:

```
python certify.py                # runs all four gates -> "SIMULATOR: CERTIFIED"
python examples/delta_demo.py    # generates the ensemble -> Delta ~ 0.09 bits
```

Both are deterministic given the seed bases. The build is documented step-by-step under `docs/` (`docs/Phase_*.md`), each with a recorded-output results note (`docs/*_Results.md`); the measurement specifically is `docs/Phase_Delta_Implementation.md` and `docs/Delta_Results.md`. Certification artifacts (gate summary, snapshot) are under `certification_output/`.

Stack. Python 3 with NumPy; pytest for the gates. No specialized hardware.

10. Conclusion

A structural evaluation of the Event Density substrate concluded that its decoupling surfaces are determinability boundaries — horizons past which one region cannot determine another. This paper turned that conclusion into a measurement. We built a faithful, certified simulator of the substrate’s selection dynamics, proved (via four pre-registered gates, 20/20 automated tests) that it monotonically accumulates, never cycles, severs reciprocal participation across decoupling surfaces, and selects uniquely. On a certified two-stratum test substrate we measured the determinability-boundary contrast at

$$\Delta \approx 0.09 \text{ bits,}$$

positive, stable, and well above the estimator’s null floor, with the across-boundary term indistinguishable from a shuffle control — exactly as the structural theory predicts.

The result is a *proof-of-method*: a finite-reach substrate’s determinability boundary is measurable in bits, with the information-theoretic controls behaving as required. The specific magnitude belongs to the test configuration, not to ED as such; the work is structural rather than physical; and the parallel to a number-theoretic boundary is one of architectural form alone. Within those bounds, the chain is complete:

The architecture established that the boundary exists. The certified simulator demonstrated that it cleanly separates the two sides. The measurement put a positive, stable number of bits on what it blocks.

A configuration-independent value — the step that would make Δ a measured property of the architecture itself — was the natural next undertaking. It was subsequently run. The postscript reports what it settled.

11. Postscript: what the robustness sweeps settled

The proof-of-method above measured Δ for one configuration and flagged the obvious next question: is the magnitude an intrinsic property of ED, or an artifact of the chosen substrate and observable? Two robustness sweeps answered it (`docs/SizeSweep_Results.md`, `docs/ObservableSweep_Results.md`).

- **Size and estimator.** The 2-bin histogram estimator made Δ *appear* to converge to ~ 0.92 bits across a $16\times$ size range — but that was an estimator artifact: varying the bin count alone swung it 0.46–0.99. A binning-free k-nearest-neighbour (KSG) estimator, stable under its own parameter, gave $\Delta \approx 1.0$ bit, stable across size 16–256.
- **Observable (the decisive test).** Holding size and estimator fixed and changing only *how the final state is read out*, Δ ranged from **0.17 to 0.98 bits** (summed $\rho \approx 0.98$; per-node vectors ≈ 0.94 ; a local window ≈ 0.17 ; the gradient ≈ 0.36), each KSG-stable. **The ~ 1 bit was the information content of the summed_rho observable — not an intrinsic property of ED.** There is no single scalar “ Δ for ED.”
- **What survived every dial.** Across-boundary mutual information $M_2 \approx 0$ held under *every* size, estimator, and observable. This is exactly what theory demands: independence is observable-invariant (the mutual information between any function of one side and any function of the other

vanishes), whereas the *strength* of dependence is observable-relative. So the magnitude was never going to be a single number — but the **zero is**.

This sharpens the paper’s central claim and the parallel to the parity barrier of multiplicative number theory. A clean two-line statement of the result:

Both systems have a boundary that destroys determination. The Factor Skyline programme quantifies its boundary with a scalar (an escape density ≈ 0.265); the Event Density substrate quantifies its boundary structurally.

The sharpening: ED’s structural quantification **is the zero**. The robust, observable-invariant statement is $M_2 = 0$ — across every readout, determination does not merely shrink across the boundary, it vanishes completely, and that total severance is the structural quantity. ED *can* produce scalars (0.17, 0.36, 0.94, 0.98); they are simply observable-relative, not canonical. “Structural” therefore does not mean “no number available” — it means **the intrinsic characterization is the invariant, not a number**.

And a hypothesis for *why* the two architectures differ in this way — offered as a conjecture, not a result, but the most interesting thing the sweeps hint at:

The Factor Skyline gets a scalar because it has a canonical observable; ED does not. FS’s multiplicative structure singles out *what to measure* — the parity / Möbius content — so “what fraction escapes determinability” has a privileged answer (≈ 0.265). ED’s substrate privileges no single function of ρ , so “how much determination crosses the boundary” has no canonical readout, and hence no canonical scalar — only the observable-independent invariant, the zero. If this is right, the scalar-versus-structural difference is not a measurement limitation but a genuine structural difference between the two architectures: **one has a preferred coordinate; the other does not**.

Testing that conjecture — whether an observable-*independent* object (a channel capacity, a transfer-entropy / causal-influence measure) recovers a canonical ED invariant where single-observable mutual information cannot — is a reformulation rather than a parameter sweep, and the natural next theoretical direction. Until then, the bounded, robust result stands: **the determinability boundary severs perfectly and observably-invariantly; its severance, not a scalar, is what characterizes it**.

References (internal corpus)

Structural evaluation (the determinability boundary, established): evaluations/ED_Substrate/ — Memo_00 (scoping); Phase_A/A1/A2 (substrate-level AD criteria and extraction modes); Phase_B-E (architecture, envelope, extremal dynamics, constraint surface; the determinability boundary located at the decoupling surface); Phase_F (criteria verdict, 6/6); Phase_G (generalization); tie-break and orientation-primitivity closures.

This measurement (the empirical arc): evaluations/ED_Substrate/Bits/ — README.md (plain-language summary) and this paper at the root; certify.py (one-command verifier); code in simulator/, analysis/ (incl. KSG estimator and observables), tests/, examples/ (incl. size_sweep.py, observable_sweep.py). The full build narrative and results notes live in docs/ — measurement plan, simulator design and implementation plan, build-step implementations and results (Milestone 1 through certification), docs/Phase_Delta_Implementation.md and

docs/Delta_Results.md (the measurement), and the robustness sweeps docs/SizeSweep_Results.md, docs/Phase_ObservableSweep_Implementation.md, docs/ObservableSweep_Results.md (§11).

Motivating note: AD_Note_SubstrateBeneathTheShadow.md — the finite-reach / determinability-boundary / projection-shadow motif shared (in form) by ED and the Factor Skyline programme.

Architectural Distillation evaluates the structural integrity of an architecture, independent of empirical interpretation. This document reports a structural measurement; it makes no claim about the physical correctness of Event Density.